
Requirements Document

Project Title: 2048 Console Game

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Overview

This project is a console-based 2048 game. The player slides tiles on a 4×4 board to merge tiles of the same number. The game ends when the player reaches the target tile (determined by the selected difficulty) or when no valid moves remain.

Requirements

1. Difficulty Selection

1. At the start of the game, the user is prompted to select a difficulty: Easy, Medium, or Hard.
2. The user selects a difficulty by entering 1 (Easy), 2 (Medium), or 3 (Hard).
3. If the user enters an invalid input, the game asks for input again.
4. The target tile for each difficulty is as follows: Easy = 256, Medium = 1024, Hard = 2048.

2. Initial State

5. The board is a 4×4 grid, initially empty.
6. After difficulty selection, two tiles are placed at random empty positions on the board.
7. Each initial tile has a value of either 2 (90% probability) or 4 (10% probability).

3. Display

8. After every move, the current board state and the current score are printed to the console.
9. Empty cells are displayed as . or 0.
10. The current difficulty and target tile value are displayed at the top of the board.

4. Input and Movement

11. The player moves all tiles by entering W (up), A (left), S (down), or D (right). Input is case-insensitive.
12. If the player enters an invalid key, the game prints an error message and asks for input again without changing the board.

13. All tiles slide as far as possible in the chosen direction.
14. If two tiles of the same value collide during a move, they merge into one tile with twice the value.
15. Each tile can be merged at most once per move.

5. Tile Generation

16. After each valid move that changes the board state, one new tile is placed at a random empty cell.
17. The new tile has a value of 2 (90% probability) or 4 (10% probability).
18. If the board does not change after a move, no new tile is generated.

6. Scoring

19. When two tiles merge, the resulting tile's value is added to the score.
20. The score starts at 0 and increases throughout the game.

7. Win Condition

21. If a tile equal to the target value is created, the game displays a win message and ends.
22. The win message includes the final score.

8. Lose Condition

23. If no empty cells remain and no two adjacent tiles (horizontally or vertically) have the same value, the game displays a game over message and ends.
24. The game over message includes the final score.